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IOT-ENABLED SMART WASTE COLLECTION SYSTEM WITH COMMUNITY SHARING AND REAL-TIME ROUTE OPTIMIZATION

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INTRODUCTION

Efficient waste collection is essential for cleanliness, sustainability, and community well-being. Challenges such as delays, missed locations, and inefficiencies can strain resources and increase costs. By incorporating real-time tracking, route optimization, and community sharing of reusable items, this system encourages timely pickups, reduces waste, and fosters collaboration among citizens to promote a cleaner, greener environment



PROBLEM DOMAIN

Current waste collection systems face significant challenges that hinder efficiency and sustainability

- **Inefficient Collection:** Delays, missed pickups, and unoptimized routes increase costs
- **No Real-Time Tracking:** Limits monitoring of bins and vehicles
- **Reusable Waste Loss:** Lack of community sharing discards reusable items
- **Environmental Risks:** Accumulated waste causes pollution and health hazards
- **Low Public Involvement:** Lack of tools for citizen participation

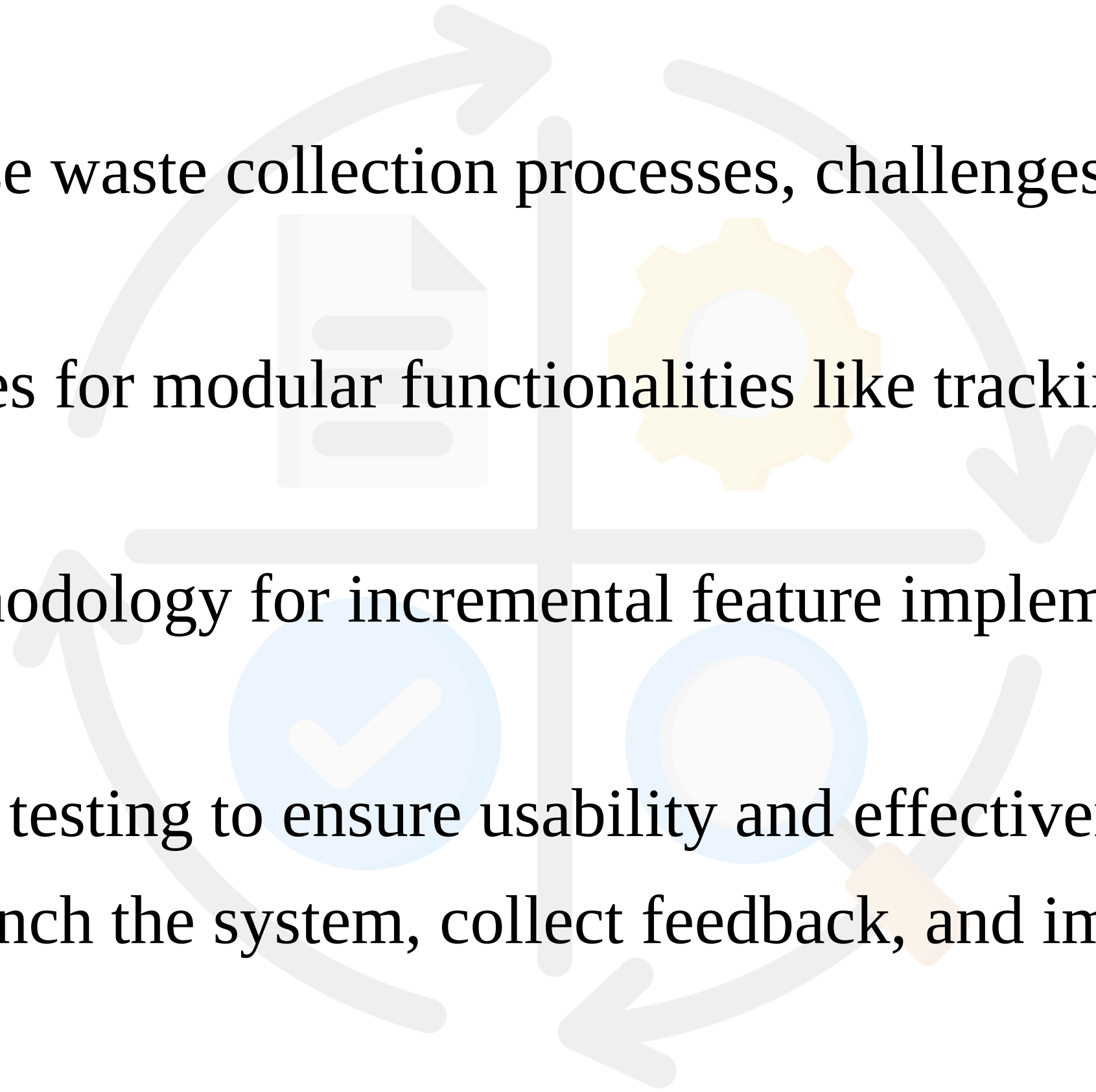
PROPOSED SOLUTION

To tackle the challenges in waste management, a comprehensive solution is proposed, integrating advanced technologies and community-driven initiatives to enhance efficiency, sustainability, and public engagement

- **IoT-enabled bins** for real-time monitoring
- **AI-based route optimization**
- **Mobile/web apps** for tracking and reporting
- **Community sharing** to reduce waste
- **Real-time truck tracking** and **data analytics**



METHODOLOGY

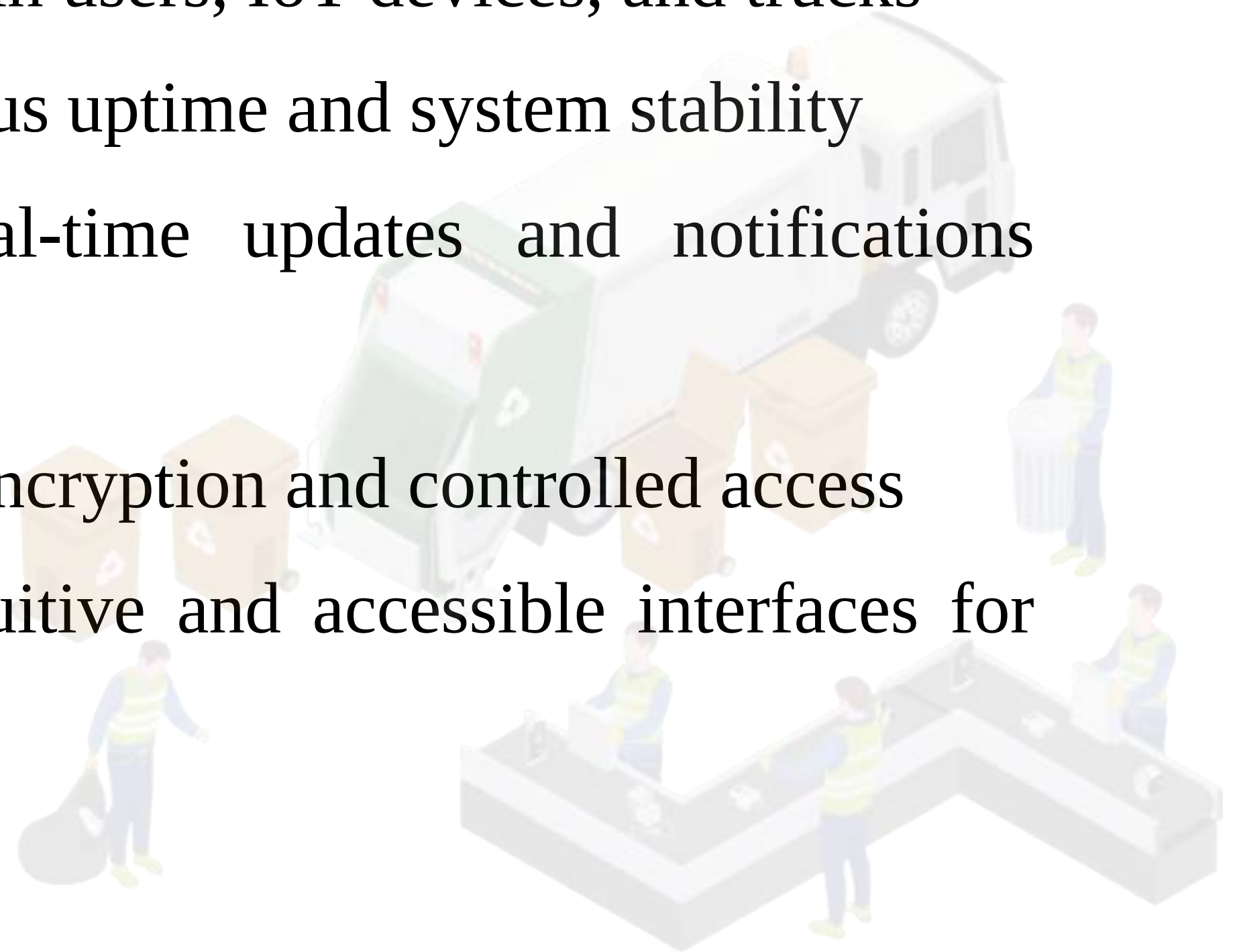
- **Requirement Gathering:** Analyze waste collection processes, challenges, and user needs with stakeholders
 - **System Design:** Use microservices for modular functionalities like tracking, notifications, and reporting
 - **Development:** Follow Agile methodology for incremental feature implementation and frequent testing
 - **Testing:** Perform user acceptance testing to ensure usability and effectiveness
 - **Deployment and Feedback:** Launch the system, collect feedback, and improve features iteratively
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FUNCTIONAL REQUIREMENTS

- **Waste Collection Monitoring:** IoT bins send real-time data; residents schedule pickups
- **Community Sharing:** Share reusable items via the app
- **Immediate Pickups:** Residents request urgent pickups; drivers get real-time alerts
- **Route Optimization & Tracking:** Generates and updates efficient routes; displays bin locations and optimized paths
- **Alerts & Notifications:** Real-time updates for users and drivers
- **Reports & Analytics:** Insights on waste trends and system performance
- **User Management:** Manage accounts, roles, and user activity

NON-FUNCTIONAL REQUIREMENTS

1. **Scalability:** Support growth in users, IoT devices, and trucks
2. **Reliability:** Ensure continuous uptime and system stability
3. **Performance:** Deliver real-time updates and notifications efficiently
4. **Security:** Protect data with encryption and controlled access
5. **User-Friendly:** Provide intuitive and accessible interfaces for all users



TECHNOLOGIES

Prototyping: Figma



Hardware (IoT Bin): ESP32, GPS (NEO-6M), LoRa/BLE, LEDs

Front-end Development: React, Flutter

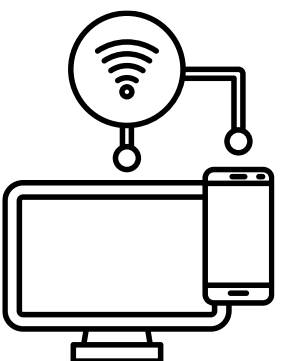
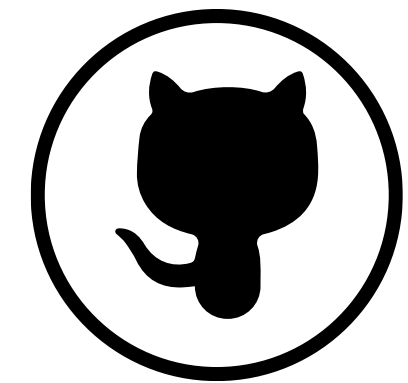
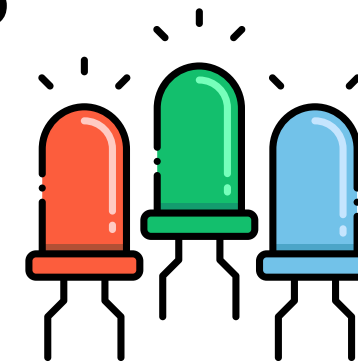
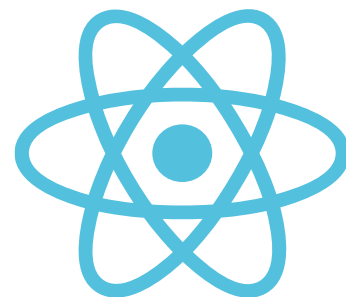
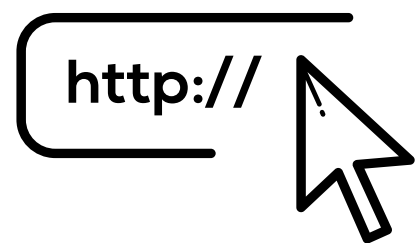


Back-end Development: Node.js, Express.js, Google Maps API

Databases: Firebase

Communication: ESP-NOW, MQTT, HTTP/HTTPS

Version Control: Git, GitHub



Literature Review - Summary

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Focus	Key Findings	Reference
IoT	IoT-enabled real-time monitoring reduced overflow risks, improved efficiency, and saved resources	Zeb et al, Haque et al., Folianto et al.
Route Optimization	Optimized collection routes, minimized carbon emissions, and reduced fuel consumption	Lozano et al., Mukherjee et al.
Cost Efficiency	Enhanced cost efficiency, reduced operational costs, and saved energy	Mukherjee et al., Lozano et al.
AI & Machine Learning	Improved prediction accuracy and reduced waste transport inefficiencies through AI algorithms	Rutqvist et al.
Community Engagement	Encouraged community participation, improved recycling rates, and supported 3R principles	Abd Wahab et al.

Focus	Key Findings	Reference
Cost Reduction & Sustainability	Reduced cleanup costs by 58 million USD and aligned with SDGs	Mager et al.
Waste Bin Monitoring	Improved waste bin filling forecasts, reduced overflow situations, and enabled efficient resource use	Morais et al.
Recycling & User Interface	Addressed recycling inefficiencies and hazardous material handling through tailored user interfaces	Donny et al.
Sustainability	Enhanced traceability and promoted sustainability through smart waste management solutions	Fatimah et al.
Energy Conversion	Highlighted energy conversion methods and cost-effective waste management strategies	Mukherjee et al.

THANK YOU